

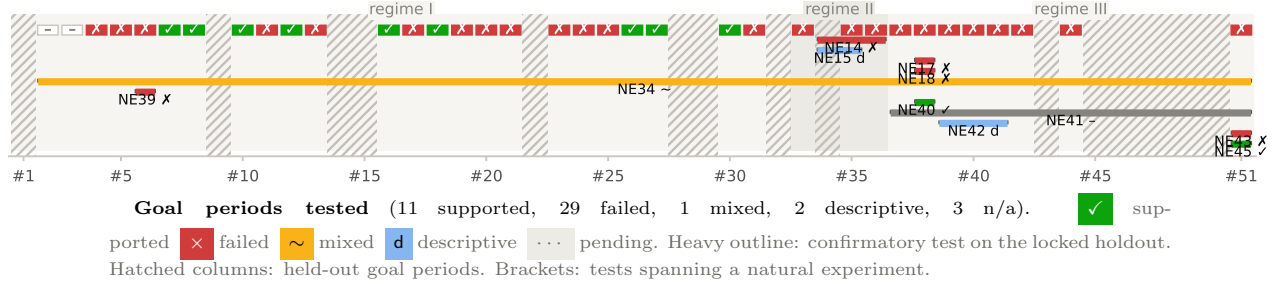
H36: A reorganization alarm: susceptibility and multi-information peak at transitions

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 65% Faithfulness 1/5 Usefulness 3/5

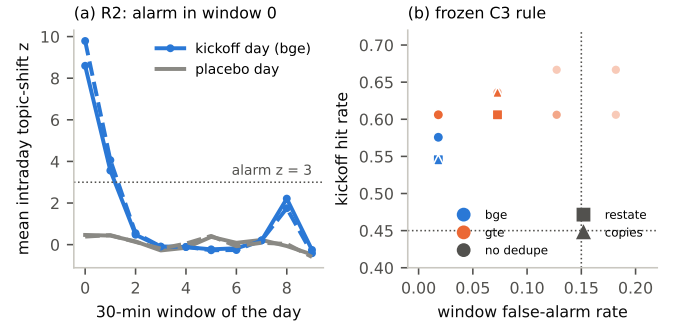
Bottom line. A plain topic-shift alarm times goal changes to the first 30 minutes of talk and survives every robustness check; the multi-information/susceptibility alarm does not beat random dates.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

H36 asks whether fluctuation statistics (multi-information, susceptibility, a heat-capacity analogue) flag swarm reorganizations. Rounds 1 and 1b: the physics alarm catches a quarter of 33 goal changes, all via content; the topic shift R1 scores AUC 0.95. **Round 2** (non-reserved data). (i) **Intraday**: a 30-min topic-shift alarm fires in the first window of day 0 for 82–85% of kickoffs, against 7–9% of placebo day starts (AUC 0.92–0.94). (ii) **Robustness**: the frozen rule C3 ($R_1 \geq 3$ or $Z_{\text{cont}} \geq 2$) keeps hit 0.55–0.64 at window FAR 0.02–0.07 in 6 embedding × dedupe variants and on Monday-matched placebos. The gte FAR moves with the surrogate seed (0.07–0.18). (iii) Sampling-invariant activity statistics still react to roster and schedule changes. (iv) An action-mix scaffold detector fails (AUC 0.61, $p = 0.16$). The physics alarm fails random dates in every variant (p 0.12–0.25).



Observables. (a) Mean intraday topic-shift z by 30-min window: kickoff days (blue; bge solid, gte dashed) against placebo days (gray). The kickoff signal sits in window 0 and is gone by window 2. (b) The frozen C3 rule: kickoff hit rate against placebo-window false-alarm rate in 6 variants (large markers) and over 10 surrogate seeds (faint dots). Dotted lines: the pre-registered robustness bounds.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An operator who wants to know that a swarm’s goal changed should watch the topic, not the fluctuations. The mean content vector moves within the first 30 minutes of talk after a kickoff, at a 4% per-window false-alarm rate. The daily rule C3 is now robust to the embedding model, the dedupe rule and the Monday calendar. With bge-small it is stable over seeds; with gte, Z_{cont} needs more surrogates. Multi-information and susceptibility add nothing beyond this: the physics alarm does not beat random dates. Platform changes need the schema diff (H74). The agents’ own day-to-day work mix is too noisy to show a prompt edit.

$$\text{VR} = \frac{\text{Var}_t M}{\sum_i \text{Var}_t s_i}, \quad I_G = -\frac{1}{2} \log \det R, \quad Z_{\text{phys}} = \langle Z_I, Z_X, Z_C \rangle \geq 2$$

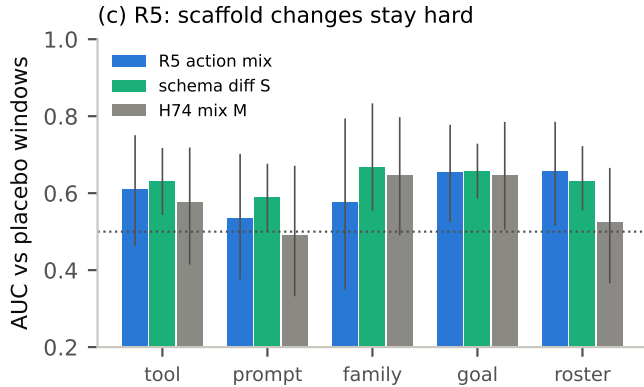
$$D(w) = 1 - \cos(\bar{\mathbf{m}}_w, \langle \bar{\mathbf{m}} \rangle_{w-4..w-1}), \quad \text{alarm: } z_{16}[D(w)] \geq 3$$

$$\text{C3: } z_{10}[R_1(d)] \geq 3 \text{ or } Z_{\text{cont}}(d) \geq 2 \text{ on } d \in \{-1, 0, 1\}$$

H36-reorganization-alarm · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H36-reorganization-alarm/README.md · ratings: H36 agent (suggested; round 2)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 (r1/1b) physics alarm on goals	AUC 0.66–0.68; random-date 0.12–0.25 in all 6 r2 variants	<i>p</i> failed
P3 rival R1 wins	AUC 0.95–0.97; vs Monday placebos 0.93–0.97	held
RB1–3 C3 robust, Monday-safe	hit 0.55–0.64; FAR 0.02–0.07; Monday FAR ≤ 0.12	held
RB4 seed-stable	bge yes; gte FAR 0.07–0.18 over 10 seeds	partly
RB5 style-resid R1	0.91 / 0.91	held
AUC ≥ 0.90		
P2.1 intraday kick-off hit	windows 0–1: 0.82 / 0.85 vs placebo 0.09 / 0.07	held
P2.3 first alarm in window 0–1	27/27, 28/28 in window 0	held
P2.4 #focus mid-day room	z 2.3 (bge), 3.2 (gte) one window early	mixed
P3.2–3 invariant activity	AUC 0.64 [0.48, 0.79]; added hits are roster/schedule	mixed
P5.1 scaffold detector	AUC 0.61 [0.46, 0.75], <i>p</i> 0.16; failed NE14b fires	



Second figure. Scaffold detection, AUC of the window maximum against placebo windows (95% CI) per event class: the action-mix detector R5, H74’s schema diff S and H74’s mix channel M. All overlap 0.5–0.7. Synthetic power: only steps larger than the agents’ own day-to-day mix noise are visible.

GOAL PERIODS IN DETAIL

G04 $\times$ 1b: failed (unchanged)
 G05 $\times$ 1b: failed (unchanged)
 G06 $\times$ 1b: failed (unchanged)
 G07 $\checkmark$ 1b: supported (bge) / failed (gte)
 G08 $\checkmark$ 1b: supported (unchanged)
 G10 $\checkmark$ 1b: supported (unchanged)
 G11 $\times$ 1b: failed (unchanged)
 G12 $\checkmark$ 1b: supported (unchanged)
 G13 $\times$ 1b: failed (unchanged)
 G16 $\checkmark$ 1b: supported (unchanged)
 +36 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Most kickoffs (26 of 33) fall on Mondays, and there are only 10 strict Monday placebos (24 in the wider set). Round 2 uses 55 placebo days, none in regime II. The intraday lag after the kickoff message is set by the schedule (the first talk window), not by swarm dynamics. The 05-04 room merge is a day-start event, so it cannot be separated from its kickoff. The #focus timing is one event. The seed study is post hoc. The activity channel’s added hits coincide with roster, room or schedule changes. The reserved data is not run.

ROUND 2

Re-freeze `confirm.py` with bge as primary and ≥ 100 surrogates for Z_{cont} ; freeze the intraday window-0 alarm for confirmation; try a presence/retirement rule on bash heads (H74-R2).

- **Essence.** a cheap, log-computable alarm for structural change in a swarm (goal, roster, room, scaffold) with known hit and false-alarm rates; round 1 says content carries it, activity fluctuations don’t, and a first-moment topic-shift detector beats the physics statistics.
- **Round-3 redirects (newest).**
- **H36-R1** (unchanged). Confirm C3 on the reserved data.
- **H36-R6** Intraday C3 on the reserved data. Freeze r1w ≥ 3 in windows 0–1 of a day as an operator alarm and confirm its hit and day-start FAR.
- **H36-R7** Bash-grammar presence rule. Combine with H74-R2’s presence/retirement rule (a head present on ≥ 8 of 10 days disappears, or a new head appears in ≥ 2 agents) to cut the tool-mix noise.